

AWS checklist and maintenance guide

Station name	TAS-L	
People present	RF + SIBA	instructed by:
Purpose	standard maintenance / installation / take-down / quick visit / emergency	
Date	11/8-2015	
Transport	helicopter / fixed wing / snow mobile / dog sleds / on foot from:	
Weather	<u>warm</u> / cold / very cold / <u>no wind</u> / some wind / rain / cloud cover %: 20	

Notes: Revisit 13/8-2015
 Scuffed logger box in MIT-1 as
 the

Metadata before maintenance (if establishing a new AWS: skip this page)

Time difference between logger clock and UTC	34 +/-
Adjust logger time to UTC (be sure PC runs on UTC time)	Y / N
Download data to PC and/or change CF Card (wait for green light)	(Y) N
Name of logger program	Promice 2014 d.
Photo or screen dump of values in fast scan mode	(Y) N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	(Y) N
Mast tilt in boom direction (+ if radiometer tilting down)	7 °
Mast tilt across boom looking from radiometer (- if clockwise)	1 °
Boom direction relative to north	10 ° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.	180° away from mast (Y) N
Radiometer aligned with mast and inclinometer? If not, measure.	(Y) N
Temperature / humidity height (bottom of casing – surface)	263 cm
Sonic ranger on AWS height (membrane – surface)	270 cm
Wind sensor height (center of propeller – surface)	312 cm
Sonic ranger on stakes height (membrane – surface)	/ cm
Free length of stake – outer, holding boom with sonic ranger	345 cm
Free length of stake – middle	323 cm
Free length of stake – outer, holding boom without sonic ranger	317 cm
Free length of ablation hose on/above the ice surface	1800 cm
Alt: depth of ablation hose using markings on hose	? cm
Height of ablation hose fluid level above the ice surface	145 cm
Vertical difference in surface height (at station – at hose) (approx)	20 cm
Length of thermistor string on surface from surface marking	600 cm
Height of surface irregularities	30 cm
Stake assembly still standing?	(Y) N / more or less
Apparent damage to sensors? If yes, which?	Y / (N)

change CR card CR 1

No markings

Booms fallen down except sonic rangers on stakes fallen down

Notes:

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	080202 61180277	800165	- Aligned with mast: Y/N - Clean: Y/N
Inclinometer	00639	SN01128	- Aligned with radiometer within 0.5°: Y/N
Satellite antenna	GEUS 28	-11-	
Wind sensor	110445	105089	- Box on sensor exactly towards mast (south): Y/N
Temperature / humidity assembly	61138896		
T/hum assembly casing			- Fan spins and sounds OK: Y/N
T/hum assembly bracket			
Temperature probe in T/hum assembly			
HygroClip in T/hum assembly	61180277	61180300 61180300	- 40 °C offset in HygroClip temperature: Y/N
Sonic ranger on AWS	Unrecoverable	1736	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N
Solar panel		C10303006 2251886	- Output OK: Y/N - Clean: Y/N
GPS antenna	GEUS 28	GEUS 22812	- Stuck to top of logger enclosure: Y/N
Data logger	AWS 2/2009	AWS 2/2009	- New internal battery in old logger: Y/N - New operating system: Y/N
Card reader		4690	- LED green or flashing orange (active): Y/N
Multiplexer		E3202	- Clicking as it should: Y/N
Iridium modem IMEI number	GEUS 28	30003401225 GEUS 28	- Transmissions tested: Y/N - Problems: Y/N
Barometer	3215015	5445946	3766881
Sonic ranger on stakes		6647	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N - In danger of being buried next winter: Y/N
Thermistor string		M02	- Old string left in ice: Y/N/cut
Ablation hose assembly		5015 ABS	- Old hose left in ice: Y/N/cut - Air in hose: Y/N/removed - Bladder 50% full: Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N AWS 2/2009		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N		

Metadata after maintenance / establishment

Time logger clock set (to UTC)?	☒ Y / ☐ N
Name of logger program	ProMile 2015e - CR1
Photo or screen dump of values in fast scan mode	☒ Y / ☐ N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	6 °
Mast tilt across boom looking from radiometer (- if clockwise)	0 °
Boom direction relative to north (radiometer should point to true south)	10 ° true/magn
Box on wind sensor exactly towards mast? If not, change.	☒ Y / ☐ N
Radiometer aligned with mast and inclinometer? If not, align.	☒ Y / ☐ N
Temperature / humidity height (bottom of casing – surface)	260 cm
Sonic ranger on AWS height (membrane – surface)	273 cm
Wind sensor height (center of propeller – surface)	323 cm
Sonic ranger on stakes height (membrane – surface)	184 cm
Free length of stake – outer, holding boom with sonic ranger	206 cm
Free length of stake – middle	220 cm
Free length of stake – outer, holding boom without sonic ranger	144 cm
Free length of ablation hose on/above the ice surface	1245 cm
Alt: depth of ablation hose using markings on hose	1255 cm
Height of ablation hose fluid level above the ice surface	140 cm
Vertical difference in surface height (at station – at hose) (approx)	10 cm
Length of thermistor string on surface from surface marking	-20 cm
Height of surface irregularities	50 cm

Notes: